

Objective

Analyze stock market data to understand price movements, trading activity, relationships between variables, and key statistical patterns.

Dataset

- 497,461 observations
- 7 columns
- Features:
 - symbol
 - date
 - open
 - high
 - low
 - close
 - volume

Key Findings

- Apple (AAPL) showed a strong upward trend in closing prices over the observed period.
- BAC recorded the highest total trading volume among the analyzed stocks.
- Open, High, Low, and Close prices are highly correlated.
- Trading volume has a weak correlation with stock prices.
- The dataset contains only a small number of missing values.
- Closing prices are right-skewed, with most stocks concentrated in lower price ranges.

Conclusion

The stock market dataset was successfully analyzed using exploratory data analysis techniques. Price-related variables (Open, High, Low, and Close) showed very strong positive correlations, indicating they move together closely. Apple (AAPL) demonstrated a strong upward trend in closing prices, while BAC recorded the highest trading volume. The dataset contained very few missing values, and the distribution of closing prices was right-skewed. Overall, the analysis provides valuable insights into stock price behavior and trading activity.

```
In [3]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

plt.style.use('ggplot')
```

```
In [6]: import os

os.getcwd()
```

```
Out[6]: 'C:\\Users\\OMEN LAPTOP 15'
```

```
In [7]: os.listdir()
```

```
Out[7]: ['-1.14-windows.xml',
        '.blip-device-dedupe-id',
        '.cache',
        '.codex',
        '.ipynb_checkpoints',
        '.ipython',
        '.jupyter',
        '.matplotlib',
        '.ms-ad',
        '.skiko',
        'AppData',
        'Application Data',
        'battery-report.html',
        'Contacts',
        'Cookies',
        'Documents',
        'Downloads',
        'Favorites',
        'Heaven',
        'inst.ini',
        'IntelGraphicsProfiles',
        'Iris_EDA_Project.ipynb',
        'Links',
        'Local Settings',
        'My Documents',
        'NetHood',
        'NTUSER.DAT',
        'ntuser.dat.LOG1',
        'ntuser.dat.LOG2',
        'NTUSER.DAT{6f212a49-ec98-11ef-81d7-af459868b8fd}.TM.blf',
        'NTUSER.DAT{6f212a49-ec98-11ef-81d7-af459868b8fd}.TMContainer000000000000000001.regtrans-ms',
        'NTUSER.DAT{6f212a49-ec98-11ef-81d7-af459868b8fd}.TMContainer000000000000000002.regtrans-ms',
        'ntuser.ini',
        'nuuid.ini',
        'OneDrive',
        'PrintHood',
        'Recent',
        'Saved Games',
        'Searches',
        'SendTo',
        'Start Menu',
        'Stock_Prices_EDA_Project',
```

```
'Templates',
'userid.ini',
'vmlogs']
```

In [8]: `import pandas as pd`

```
df = pd.read_csv("Downloads/2) Stock Prices Data Set.csv")
df.head()
```

Out[8]:

	symbol	date	open	high	low	close	volume
0	AAL	2014-01-02	25.0700	25.8200	25.0600	25.3600	8998943
1	AAPL	2014-01-02	79.3828	79.5756	78.8601	79.0185	58791957
2	AAP	2014-01-02	110.3600	111.8800	109.2900	109.7400	542711
3	ABBV	2014-01-02	52.1200	52.3300	51.5200	51.9800	4569061
4	ABC	2014-01-02	70.1100	70.2300	69.4800	69.8900	1148391

In [9]: `print("shape:", df.shape)`
`df.info()`

```
shape: (497472, 7)
<class 'pandas.DataFrame'>
RangeIndex: 497472 entries, 0 to 497471
Data columns (total 7 columns):
#   Column  Non-Null Count  Dtype
---  -
0   symbol  497472 non-null    str
1   date    497472 non-null    str
2   open    497461 non-null    float64
3   high    497464 non-null    float64
4   low     497464 non-null    float64
5   close   497472 non-null    float64
6   volume  497472 non-null    int64
dtypes: float64(4), int64(1), str(2)
memory usage: 26.6 MB
```

In [10]: `df.isnull().sum()`

```
Out[10]: symbol      0  
         date        0  
         open       11  
         high        8  
         low         8  
         close       0  
         volume      0  
         dtype: int64
```

```
In [11]: df.describe()
```

```
Out[11]:
```

	open	high	low	close	volume
count	497461.000000	497464.000000	497464.000000	497472.000000	4.974720e+05
mean	86.352275	87.132562	85.552467	86.369082	4.253611e+06
std	101.471228	102.312062	100.570957	101.472407	8.232139e+06
min	1.620000	1.690000	1.500000	1.590000	0.000000e+00
25%	41.690000	42.090000	41.280000	41.703750	1.080166e+06
50%	64.970000	65.560000	64.353700	64.980000	2.084896e+06
75%	98.410000	99.230000	97.580000	98.420000	4.271928e+06
max	2044.000000	2067.990000	2035.110000	2049.000000	6.182376e+08

```
In [13]: df.isnull().sum()
```

```
Out[13]: symbol      0  
         date        0  
         open       11  
         high        8  
         low         8  
         close       0  
         volume      0  
         dtype: int64
```

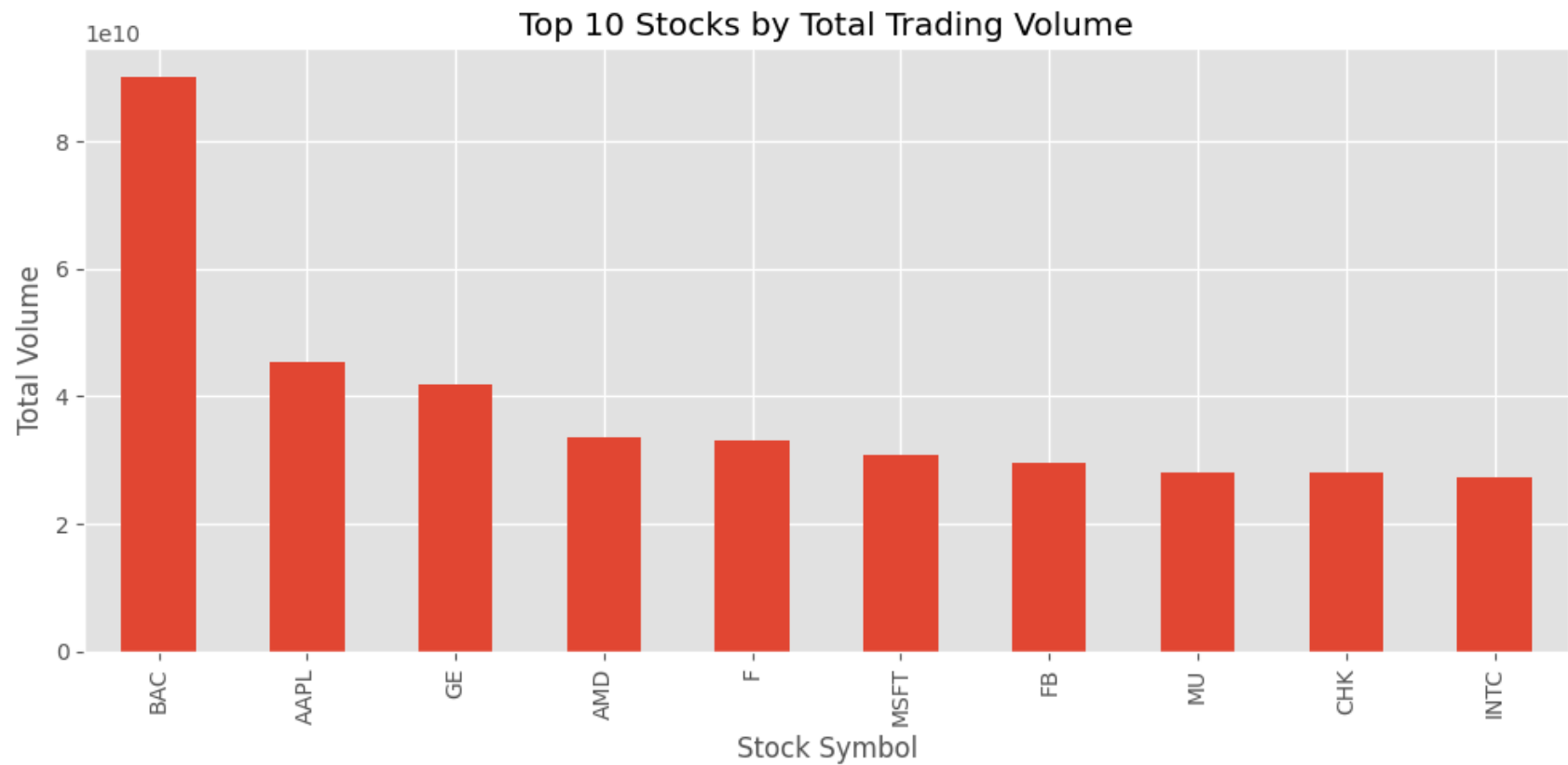
```
In [16]: top_volume = df.groupby("symbol")["volume"].sum().sort_values(ascending=False).head(10)
```

```
In [17]: import matplotlib.pyplot as plt

top_volume.plot(kind='bar', figsize=(10,5))

plt.title('Top 10 Stocks by Total Trading Volume')
plt.xlabel('Stock Symbol')
plt.ylabel('Total Volume')
plt.tight_layout()

plt.show()
```



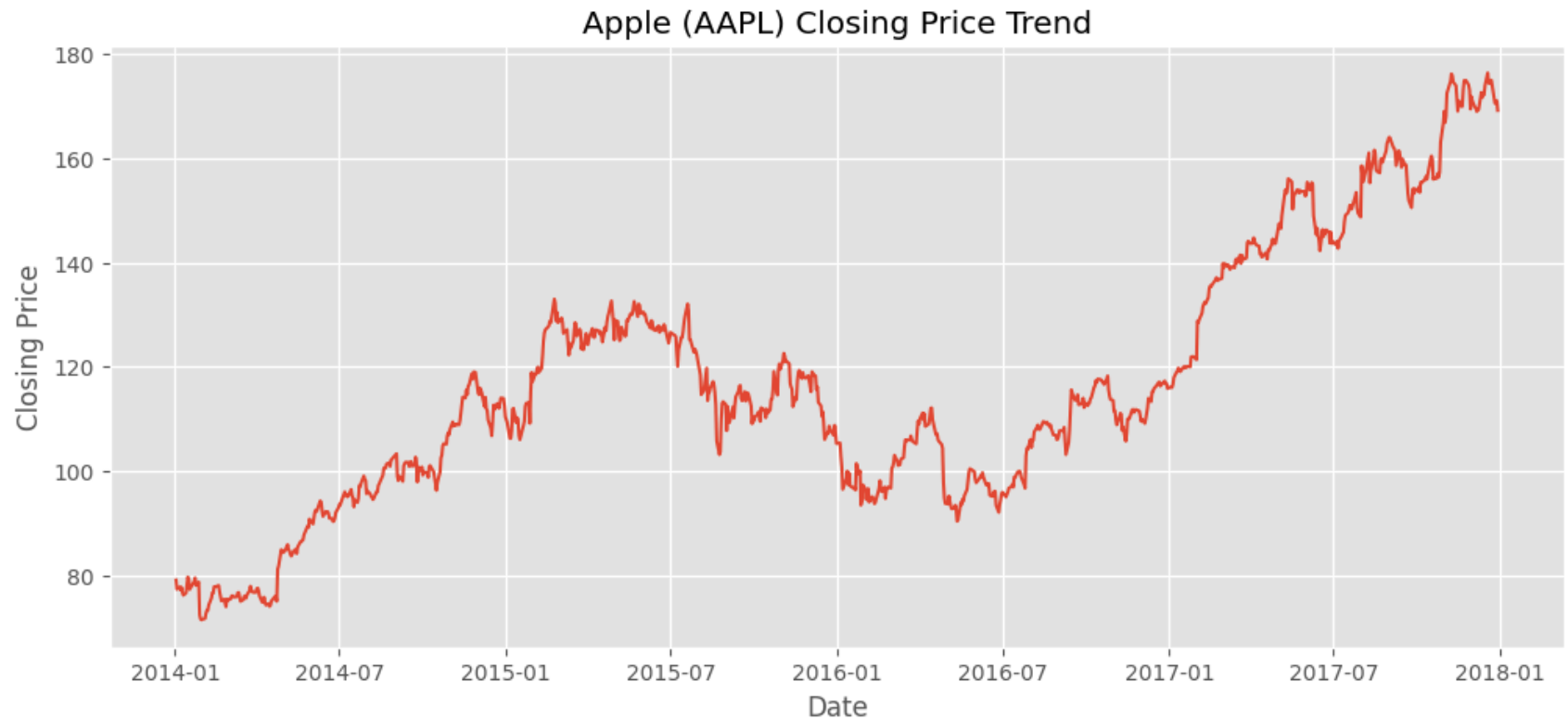
```
In [18]: df['date'] = pd.to_datetime(df['date'])
```

```
In [19]: apple = df[df['symbol'] == 'AAPL']
```

```
apple = apple.sort_values('date')

plt.figure(figsize=(12,5))
plt.plot(apple['date'], apple['close'])

plt.title('Apple (AAPL) Closing Price Trend')
plt.xlabel('Date')
plt.ylabel('Closing Price')
plt.show()
```



```
In [20]: df[['open', 'high', 'low', 'close', 'volume']].corr()
```

Out[20]:

	open	high	low	close	volume
open	1.000000	0.999941	0.999928	0.999876	-0.148724
high	0.999941	1.000000	0.999905	0.999938	-0.148344
low	0.999928	0.999905	1.000000	0.999941	-0.149284
close	0.999876	0.999938	0.999941	1.000000	-0.148839
volume	-0.148724	-0.148344	-0.149284	-0.148839	1.000000

```
In [21]: import matplotlib.pyplot as plt

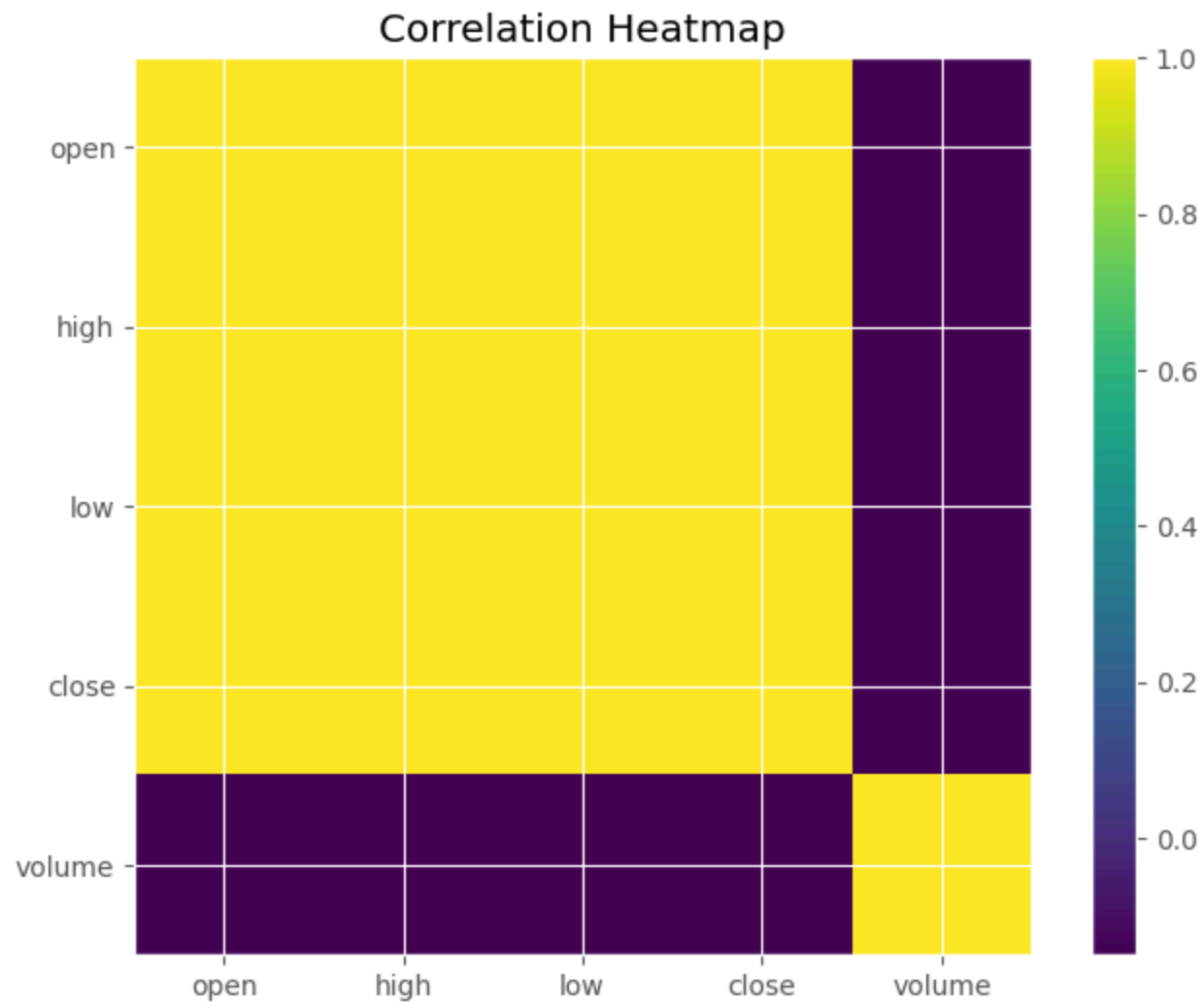
corr = df[['open', 'high', 'low', 'close', 'volume']].corr()

plt.figure(figsize=(8,6))
plt.imshow(corr)

plt.colorbar()

plt.xticks(range(len(corr.columns)), corr.columns)
plt.yticks(range(len(corr.columns)), corr.columns)

plt.title('Correlation Heatmap')
plt.show()
```

```
In [22]: df.isnull().sum()
```

```
Out[22]: symbol      0  
         date        0  
         open       11  
         high        8  
         low         8  
         close       0  
         volume      0  
         dtype: int64
```

```
In [23]: import matplotlib.pyplot as plt  
  
df['close'].hist(bins=50, figsize=(10,5))  
plt.title('Distribution of Closing Prices')  
plt.xlabel('Closing Price')  
plt.ylabel('Frequency')  
plt.show()
```



```
In [24]: top_close = df.groupby('symbol')['close'].mean().sort_values(ascending=False).head(10)

top_close.plot(kind='bar', figsize=(10,5))
plt.title('Top 10 Stocks by Average Closing Price')
plt.ylabel('Average Closing Price')
plt.show()
```



In []: